

1. The following matching: [A,B|C] with [1,b] end with:

R: Succes

2. The following matching: [1,2,3] with [A,B|C] would provide the folowing instantiations:

R: **A** = 1, B = 2, C = [3]

3. Consider the predicate having a recursive clause as the one below:

```
p([H|T], Some_Result, End_Result):-  
    combines(H, Something, New_Some_Result),  
    p(T, New_Some_Result, End_Result).
```

Using this strategy, the result would have to be initialized:

R: When the initial call is made, in the second argument

4. The recursive clause below of some predicate is designed with **backward recursion** because:

```
p([H|T],Some_Result):-  
    p(T, Some_Part_Result),  
    combines(H, Some_Part_Result,Some_Result).
```

?R: It processes the current element after ending the recursion

5. Consider a predicate having a recursive clause as the one below:

```
p([H|T], Some_Result, End_Result):-  
    combines(H, Something, New_Some_Result),  
    p(T, New_Some_Result, End_Result).
```

Using this strategy, the result "increases":

?R:As each recursive call returns, in the second argument

6. Consider a predicate having a recursive clause, as below:

```
p([H|T], Some_Result):-  
    p(T, Some_Part_Result),  
    combines(H, Some_Part_Result, Some_Result).
```

On the stopping condition:

R: The result has to be made available at the top level (call level)

7. A cut NEVER affects:

R: Subtrees of subgoals to its right

8. Whats is wrong with the following unification?

```
?- n(c(a,B), v, m(1,[2,B])) = N(c(a,B), V, m(1,[2|[3]])).?
```

R: The name of the structure on the right is not constant

9. A cut would freeze (prevent from backtracking) some nodes in the tree. However, it is NOT freezing:

R: A subtree of its sibling node to its left BEFORE the cut is executed

R: A sibling node to its left BEFORE the cut is executed

10. Which categories of nodes are frozen (prevented from backtracking) by a cut:

R: Any sibling node to its left after the execution of the cut

R: Its parent node after the execution of the cut

R: Any subtree of a sibling node to its left after the execution of the cut

11. The following matching: [A,b|C] with [1,2,3,4] fails because:

R: Both b and 2 are constants (and different)

12. When matching the following: $[A, B | C]$ with $[X, Y, Z]$ in return we get:

R: $A = X$, $B = Y$, $C = [Z]$

13. The following matching: $f(A, g(B), 4)$ with $f(a, g, 4)$ would fail because:

R: Because of the second argument ($g(B)$ is an expression while g is a constant)

14. Which one of the following clause heads will the query: $?- \text{process}([a, b, c], \text{variable}, \text{Constant})$. **NOT** match?

R: $\text{process}([A, B, C], C, v):-$

15. When matching $[X | Y]$ with $[1, 2, b]$ the result is:

R: Success, $X = 1$, $Y = [2, b]$

16. Which of the following heads will the query: $?- \text{process}([a, b, c], \text{variable}, \text{Constant})$. **Match successfully?**

R: $\text{process}([a | T], v, C):-$

17. The recursive clause below of some predicate is designed with **forward recursion** because:

```
p([H|T], Some_Result):-  
    combines(H, Something, New_Some_Result),  
    p(T, New_Some_Result).
```

?R: It processes the current element before entering in recursion